

# Learn How to Build PyTorch Neural Networks from Scratch

## Overview

Neural Networks are a set of algorithms based on the human brain, and they are the heart of deep learning algorithms. Pytorch is one of the deep learning libraries to build neural networks. Pytorch is quite pythonic, which makes building neural networks pretty simple. Pytorch can help solve business problems such as classification, text recognition, image segmentation, etc.

As a part of the Pytorch project series, this project gives a brief introduction to Neural networks and how to build them in Pytorch. We will build a neural network sequentially for a classification problem in Pytorch and understand how neural networks are trained. If you haven't already visited, here is the first project of the series [Hands-On Approach to Master PyTorch Tensors with Examples](#).

## Aim

- To understand Neural networks
- To classify if a customer will churn or not by using a neural network in Pytorch

## Data Description

The dataset used in this project has information about the customer churn based on the various features. The dataset contains 2000 rows and 15 features to predict the churn.

## Tech Stack

- Language: Python
- Libraries: pandas, pytorch, matplotlib, sci-kit learn, numpy

## Approach

- Data cleaning
- Data preprocessing
- Building a sequential neural network
- Model training

## Modular Codes Overview:

Input

|\_data.csv

Notebook

|\_NeuralNetwork.ipynb

src

|\_Engine.py

|\_ML\_Pipeline

    |\_Preprocessing.py

    |\_Pytorch\_NN.py

Readme.md

requirements.txt

Once you unzip the Pytorch\_neural\_network.zip file, you can find the following folders within it.

1. Input
  2. Notebook
  3. src
  4. Readme.md
  5. requirements.txt
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1. The input folder contains the data that we have for analysis. In our case, it contains data.csv
  2. The Notebook folder contains the jupyter notebook file of the project.
  3. The src folder is the heart of the project. This folder contains all the modularized code for all the above steps in a modularized manner. It further includes the following.
    - a. ML\_pipeline
    - b. Engine.py

The ML\_pipeline is a folder that contains all the functions put into different python files, which are appropriately named. These python functions are then called inside the Engine.py file.

4. The requirements.txt file has all the required libraries with respective versions.  
Kindly install the file by using the command **pip install -r requirements.txt**
5. **All the instructions for running the code are present in Readme.md file**

### **Takeaways**

1. What is Pytorch?
2. Features of Pytorch
3. What are the advantages of Pytorch?
4. Difference between Pytorch and Tensorflow
5. What are Neural Networks?
6. What is Node in a Neural Network?
7. Type of Nodes in Neural Network
8. What is Sequential in a Neural Network?
9. The architecture of Neural Network in Pytorch
10. How to build Neural Networks in Pytorch?
11. How are sequential layers built-in Pytorch?
12. How are Neural Networks trained in Pytorch?
13. How to convert data into Tensors?
14. Label encoding of categorical variables
15. Scaling numerical features
16. The optimizer in Neural Networks